

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	July 11 2026
Team ID	SWTID-2026-2791
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High	Team
Sprint-1	Weather Monitoring	USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High	Team
Sprint-2	Soil Monitoring	USN-3	As a user, I can register for the application through Facebook	2	Low	Team
Sprint-1	Crop Recommendation	USN-4	As a user, I can register for the application through Gmail	2	Medium	Team
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High	Team
Sprint-3	Dashboard	USN-6	As a farmer, I can view my farm dashboard with crop and weather information.	1	High	Team
Sprint-4	Irrigation Optimization	USN-7	As a farmer, I can view my farm dashboard with crop and weather information.	2	Medium	Team
Sprint-3	Disease Detection	USN-8	As a farmer, I can upload crop images to detect diseases using AI.	2	Low	Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	7 Days	01 Aug 2026	07 Aug 2026	20	07 Aug 2026
Sprint-2	18	7 Days	08 Aug 2026	14 Aug 2026	18	14 Aug 2026
Sprint-3	18	7 Days	15 Aug 2026	21 Aug 2026	18	21 Aug 2026
Sprint-4	16	7 Days	22 Aug 2026	28 Aug 2026	16	28 Aug 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

- **Total Story Points: 72**
- **Total Sprint Duration: 28 Days**
- **Average Velocity = $72 \div 28 \approx 2.57$ Story Points per Day**
- **Average Sprint Velocity = 18 Story Points per Sprint**

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

1. F. M. Talaat, "Crop yield prediction algorithm (CYPA) in precision agriculture based on IoT techniques and climate changes," *Neural Computing and Applications*, vol. 35, pp. 17281–17292, 2023.
2. S. Saha, O. D. Kucher, A. O. Utkina, and N. Y. Rebouh, "Precision agriculture for improving crop yield predictions: A literature review," *Frontiers in Agronomy*, vol. 7, 2025.
3. R. N. V. Jagan Mohan, P. S. Rayanoothala, and R. Praneetha Sree, "Next-gen agriculture: Integrating AI and XAI for precision crop yield predictions," *Frontiers in Plant Science*, vol. 15, 2025.
4. M. Ramasamy and T. K., "Integration of AI and IoT for Yield Optimization in Precision Farming," *Journal of Experimental Agriculture International*, vol. 46, no. 4, pp. 1–15, 2025.
5. Crop yield prediction using machine learning: An extensive and systematic literature review," *Agricultural Technology*, vol. 10, Article 100718, 2025.